

Student Answer Sheet

Subject: Artificial Intelligence

Definition:

- AI = Artificial Intelligence - Simulation of human intelligence by machines - Includes: learning, reasoning, self-correction - Tasks: speech recognition, visual perception, decision-making

Body:

- Machine Learning: learns from data without explicit programming - Deep Learning: multi-layer neural networks for pattern recognition - NLP: machines understand and generate language - Computer Vision: interprets images and video - Applications: * Healthcare: diagnostics * Finance: fraud detection * Transport: self-driving cars - Techniques: supervised, unsupervised, reinforcement learning

Conclusion:

- AI is transformative and reshapes industries - Benefits: automation, efficiency, decision-making - Concerns: privacy, bias, job displacement - Need: transparent, fair, accountable AI development